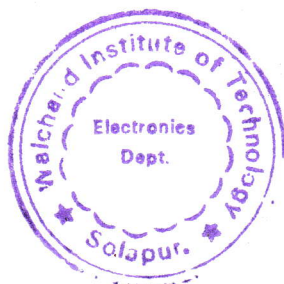


Walchand Institute of Technology, Solapur
(An Autonomous Institute)

Affiliated to
Punyashlok Ahilyadevi Holkar Solapur University,
Solapur

Choice Based Credit System (CBCS)
Structure and Syllabus
Multidisciplinary Minor Program in
Electronics and Computer Engineering
offered by
Department of Electronics Engineering
for Batch Admitted in
2024-25


Dean (Academics)



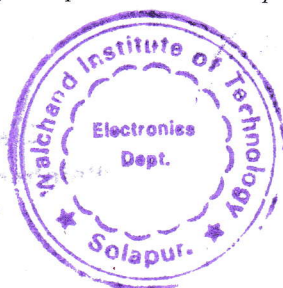

HEAD

Electronics Department
Walchand Institute of Technology
SOLAPUR-413006.

Multidisciplinary Minor Program in Electronics and Computer Engineering

Legends Used

L	Lecture Hours / week
T	Tutorial Hours / week
P	Practical Hours / week
FA	Formative Assessment
SA	Summative Assessment
ESE	End Semester Examination
ISE	In Semester Evaluation
ICA	Internal Continuous Assessment
POE	Practical and Oral Exam
OE	Oral Exam
MOOC	Massive Open Online Course
HSS	Humanity and Social Science
NPTTEL	National Programme on Technology Enhanced Learning
F.Y.	First Year
S.Y.	Second Year
T.Y.	Third Year
B.Tech.	Bachelor of Technology



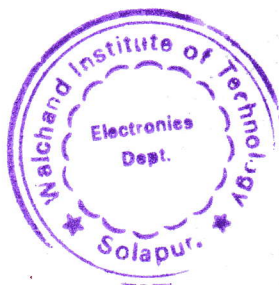
Multidisciplinary Minor Program in Electronics and Computer Engineering

Course Code Format

2	4	E	C	U/P	2	C	C	1	T/L
Year of syllabus revision	Program Code	U-Under Graduate, P-Post Graduate	Semester No./ Year1/2/3/...8	Course Type	Course Serial No. 1-9	T-Theory, L-Lab session A-Tutorial P-Programming / Design			

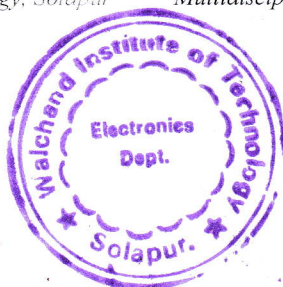
Program Code

ET	Electronics and Telecommunication Engineering
Course Type	
BS	Basic Science
ES	Engineering Science
HU	Humanities & Social Science
MC	Mandatory Course
CC	Programme Core Compulsory Course
SN*	Self-Learning (N* indicates the serial number of electives offered in the respective category)
EN*	Programme Core Elective Course (N* indicates the serial number of electives offered in the respective category)
SK	Skill-Based Course
SM	Seminar
MP	Mini project
PR	Project
IN	Internship
ON*	Open Elective (N* indicates the serial number of electives offered in the respective category)
MD	Multidisciplinary Minor



EM	Entrepreneurship/Economics/Management Courses
FP	Community Engagement Project / Field Project
AE	Ability Enhancement Course
VE	Value Education Course
IK	Indian Knowledge System
VS	Vocational & Skill Enhancement Course
RM	Research Methodology
HN	Honors' Degree Course
HR	Honors Research

Sample Course Code	
24ECU3MD6T	Analog Electronics

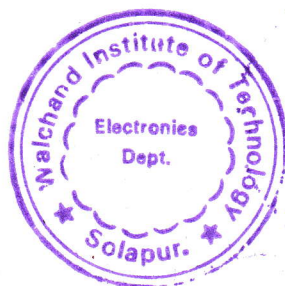


Multidisciplinary Minor Program in Electronics and Computer Engineering

Semester	Course Code	Name of Course	Engagement Hours			Credits					Total
			L	T	P		Theo-ry	OE/POE	ISE	ICA	
III	24ECU3MD6T	Analog Electronics	2	-	-	2	60	-	40	-	100
IV	24ECU4MD6T	Digital Electronics	1	-	-	1		-	50	-	50
V	24ECU5MD6T	Microcontrollers and Peripherals	3	-	-	3	60	-	40	-	100
VI	24ECU6MD6T	Advanced Controllers and Interfacing	2	-	-	2	60	-	40	-	100
VII	24ECU7MD6T	Electronic System Design	3	-	-	3	60	-	40	-	100
VII	24ECU7MD6A	Electronic System Design(Tutorial)	-	1	-	1	-	-	-	25	25
Sub-Total			11	1	0	12	240		210	25	475
Laboratory Courses											
IV	24ECU4MD6L	Digital Electronics	-	-	2	1	-	-	-	25	25
VI	24ECU6MD6L	Advanced Controllers and Interfacing	-	-	2	1	-	-	-	25	25
Sub-Total			0	0	4	2	0		0	50	50
Grand-Total			11	1	4	14	240		210	75	525

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**WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR****(An Autonomous Institute)****Third Year B.Tech. (Multidisciplinary Minor in ECM Engineering), Semester-V****24ECU5MD6T : Multidisciplinary Minor III –
Microcontrollers and Peripherals**

Teaching Scheme		Examination Scheme	
Lectures	3 Hours/week	ESE	60 Marks
Credits	3	ISE	40 Marks

Introduction:

This course introduces microcontrollers, their architecture, programming, and peripheral interfacing. It provides fundamental knowledge essential for embedded systems and real-world applications. The course is designed to be accessible to students by explaining concepts in a structured and simplified manner.

Course Prerequisite:

Students should have a basic understanding of digital electronics, logic gates, and basic C programming. Also Knowledge of Basics of Analog Electronics will help in better understanding of peripherals.

Course Objectives:

1. To introduce the basic concepts of microprocessors and microcontrollers.
2. To understand the internal architecture and working of the 8051 microcontroller.
3. To learn basic programming and interfacing techniques.
4. To explore communication protocols and real-world applications.

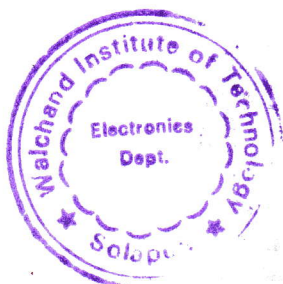
Course Outcomes:

After completing this course, student will be able to –

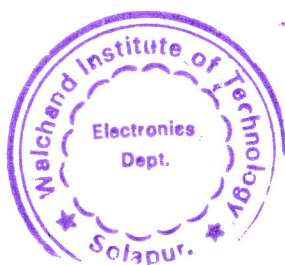
1. Differentiate between a microprocessor and a microcontroller.
2. Read and analyse the architecture and working of the 8051 microcontroller from datasheet
3. Write simple programs in Embedded C for microcontroller applications.
4. Interface LEDs, motors, and displays with a microcontroller.
5. Differentiate between communication protocols like RS232, I2C, and SPI.

Unit – I	Introduction to Microprocessors and Microcontrollers	08Hours
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Basics of microprocessors: working principles, Address, data, and control buses and their role in data processing, Memory organization: RAM, ROM, Flash, EEPROM – types and classification, Differences between microprocessors and microcontrollers, Applications of microcontrollers in automation, robotics, and consumer electronics.



Unit – II	The 8051 Microcontroller Architecture	09 Hours
Overview of 8051 microcontroller: features and applications, Internal architecture: ALU, Boolean processor, oscillator, Registers: General-purpose and special function registers, Stack, stack pointer, and program counter, Clock and reset circuits and their importance, Memory organization: program memory vs. data memory, Pin configuration and I/O port operations, Addressing modes and instruction set overview.		
Unit – III	Microcontroller Programming	09 Hours
Basics of Embedded C programming and its advantages over assembly, GPIO (General-Purpose Input / Output) and port programming, Timers and counters: operation and applications, Interrupts and their role in multitasking, Serial communication basics and UART operation.		
Unit – IV	Peripherals	08 Hours
Interfacing LEDs and Seven-segment displays, LCD interfacing: character and graphical displays, Keypad and switch interfacing techniques, Basics of motor control: DC motor and Stepper motor interfacing, Practical applications of peripheral interfacing in embedded systems.		
Unit – V	Communication Protocols	08 Hours
Serial communication: RS232, I2C and SPI communication, Interfacing external memory with microcontrollers.		
Textbooks:		
1. Muhammad Ali Mazidi, "The 8051 Microcontroller and Embedded Systems," Pearson. 2. Kenneth J. Ayala, "The 8051 Microcontroller," Thomson.		
Reference books:		
1. Raj Kamal, "Embedded Systems: Architecture, Programming and Design," McGraw Hill. 2. Frank Vahid, Tony Givargis, "Embedded System Design," Wiley.		



**WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR****(An Autonomous Institute)****Third Year B. Tech. (Multidisciplinary Minor in ECM Engineering), Semester-VI****24ECU6MD6T Multidisciplinary Minor IV-
Advanced Controllers and Interfacing**

Teaching Scheme		Examination Scheme	
Lectures	2 Hours/week	ESE	60 Marks
Practical	2 Hours/week	ISE	40 Marks
Credits	3	ICA	25 Marks

Introduction:

This course introduces advanced microcontrollers like Arduino and ESP32, focusing on their architecture, programming, and real-world interfacing applications. It emphasizes hands-on learning, enabling students to develop embedded solutions using sensors, actuators, and communication protocol.

Course Prerequisite:

Students should have completed Microcontrollers and Peripherals and have basic knowledge of embedded programming.

Course Objectives:

1. To introduce the architecture and programming of Arduino and ESP32 microcontrollers.
2. To interface various sensors, actuators, and displays with microcontrollers.
3. To explore different communication protocols in systems.
4. To implement application based on sensor-actuator based applications using ESP32.

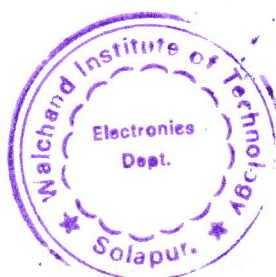
Course Outcomes:

After completing this course, student will be able to –

1. Differentiate between architectures and list features of Arduino and ESP32.
2. Write and debug embedded programs using IDE.
3. Interface sensors with microcontrollers.
4. Interface Actuators with microcontrollers.
5. Simulate Communication Protocols like RS232, I2C, SPI

Unit – I	Introduction to Advanced Microcontrollers	04 Hours
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Overview of Arduino and ESP32 microcontrollers, Comparison between Arduino (AVR) and ESP32 (Xtensa) architectures, Development environments: Arduino IDE, PlatformIO, Power management and low-power modes in ESP32.



Unit – II	Digital and Analog Interfacing	05 Hours
GPIO programming: Digital input and output, Pulse Width Modulation (PWM) and servo motor control, Analog input with ADC: Interfacing potentiometers and sensors.		
Unit – III	Sensor Interfacing	05 Hours
Interfacing temperature, humidity, and motion sensors, Ultrasonic sensor-based distance measurement. LDR Interfacing.		
Unit – IV	Actuator Interfacing	05 Hours
Stepper and DC motor control using motor drivers, Relay module interfacing for automation, LCD Display Interfacing.		
Unit – V	Communication Protocols	06 Hours
Serial communication: UART, Software Serial, I2C protocol: Interfacing multiple devices (OLED, EEPROM), SPI communication: Interfacing SD cards and displays, Bluetooth and BLE in ESP32.		
Internal Continuous Assessment (ICA):		
ICA shall consist of minimum 8 experiments based upon-		
1. Blinking LED and controlling GPIO using Arduino and ESP32 2. Interface DHT Sensor to Read Temperature and Humidity. 3. Reading analog sensors like LDR and displaying data. 4. PWM control of LED brightness and servo motor positioning 5. Interfacing Ultrasonic Sensor to Measure Distance of Object. 6. Interfacing LCD/OLED display for sensor visualization 7. UART-based serial communication between Arduino/ESP32 and PC 8. I2C communication with peripherals (OLED, RTC module, etc) 9. Relay Module Interfacing to Control HV Appliances 10. DC Motor Interfacing with ESP32 using Driver Circuit		
Textbooks:		
1. Programming Arduino: Getting Started with Sketches by Simon Monk, McGraw Hill. 2. Internet of Things with Arduino and ESP32 by Pradeeka Seneviratne, Packt.		
Reference Books:		
1. Arduino Cookbook by Michael Margolis, O'Reilly Media. 2. Beginning Sensor Networks with Arduino and Raspberry Pi, Charles Bell, Apress.		

